Topic 13: Error Handling, Retries & Recovery Strategies

n8n Hands-On Intermediate Assessment - Level 2

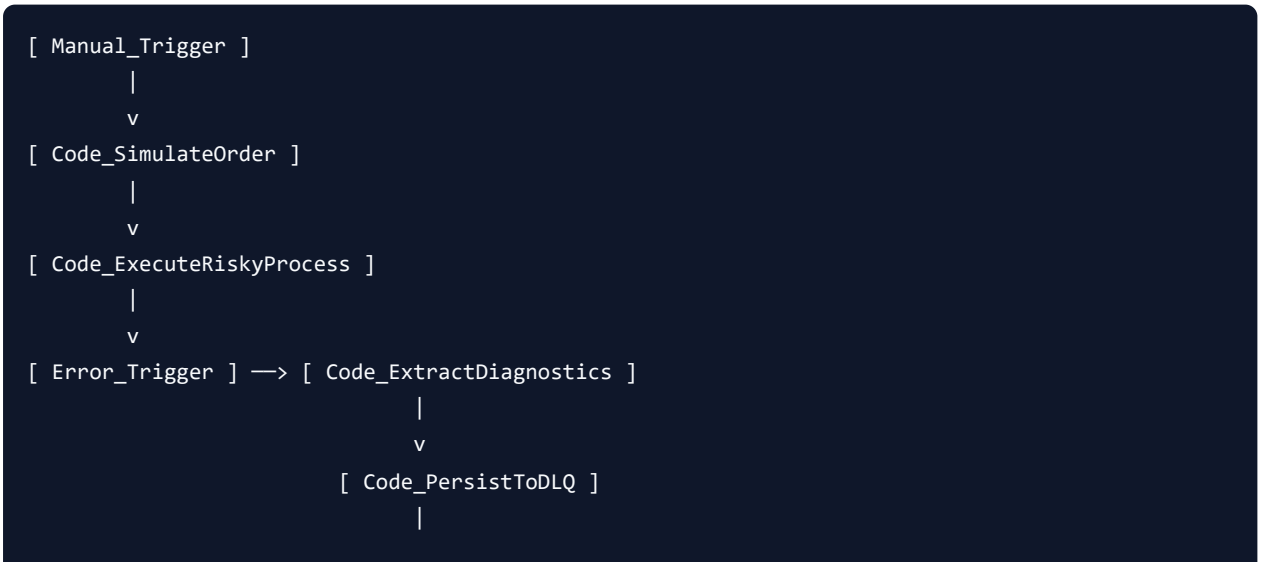
1. Document Control

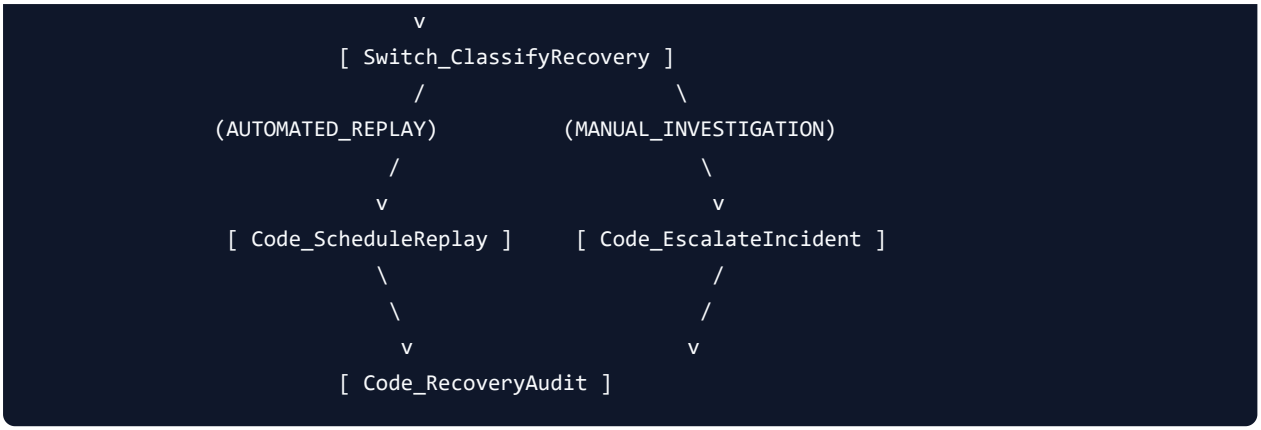
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Target Audience	Automation Engineers, Reliability Engineers, DevOps Architects

2. Executive Summary & Objective

Build an enterprise Dead Letter Queue (DLQ) and self-healing recovery pipeline in n8n. The system captures fatal execution crashes using the Error Trigger node, extracts diagnostic context and raw payload data, persists incidents into a DLQ quarantine store, bifurcates errors into Transient vs. Permanent categories, automatically replays recoverable transactions, and escalates poison-pill errors to engineering teams.

3. Architecture Diagram





4. Node Configuration Blueprint

Node Name	Type	Operation / Setting	Output Outcome
Manual_Trigger	Manual Trigger	Trigger execution on demand	Execution pulse
Code_SimulateOrder	Code	Emits transaction with fault scenario	Order test payload
Code_ExecuteRiskyProcess	Code	Simulates retries & gateway timeout	Preserved failure state
Error_Trigger	Error Trigger	Global crash listener	Execution error context
Code_ExtractDiagnostics	Code	Extracts node, error code, telemetry	Parsed diagnostic incident
Code_PersistToDLQ	Code	Stores to DLQ with unique tracking ID	dlqId assigned record
Switch_ClassifyRecovery	Switch	Classifies recovery category	Directs recovery flow
Code_ScheduleReplay	Code	Calculates backoff delay (10s)	Replay task scheduled
Code_EscalateIncident	Code	Formats high-priority SRE alert	Escalated incident card
Code_RecoveryAudit	Code	Master recovery telemetry receipt	Confirmed zero data loss

5. Test Matrix & Verification Scenarios

Scenario	Failure Mode	Classification	Expected Outcome
Transient Timeout	HTTP 504 / ETIMEDOUT	AUTOMATED_REPLAY	Stored in DLQ, routed to Code_ScheduleReplay, 10s backoff scheduled.

Permanent Schema Error	HTTP 422 / VALIDATION_ERROR	MANUAL_INVESTIGATION	Quarantined in DLQ, routed to Code_EscalateIncident, flagged "Do not retry".